Carbo_Lens — Guide Meeting Sheet

2026-09-22 · @Someone

What we are building and why

Carbo_Lens watches forest and mangrove carbon projects from satellites and warns early, with a Low / Medium / High risk level and a plain reason, when stored carbon may be in danger.

Carbon projects plant or protect trees so they store carbon, and companies pay for that storage. A fire, drought, clearing or cyclone can release it again, which is called a permanence risk. Checking sites on the ground is slow and expensive, so it happens rarely.

The risk is already recognised by the market. Carbon registries such as Verra make projects hold back a share of their credits in a buffer pool, sized with their AFOLU Non-Permanence Risk Tool. Our system gives continuous, satellite-based evidence for that same risk, using only free data.

Where the data comes from

All data is free and public, and we access it through Google Earth Engine (registered, non-commercial Community tier). Fire data can also be pulled directly from NASA with our own API key.

Data	Source	What it gives us	Detail
Satellite images	Sentinel-2, European Space Agency	Plant greenness and water, every ~5 days	10 m pixels
Fire	NASA FIRMS (MODIS / VIIRS)	Where and when fires were detected	Within hours
Forest loss	Hansen / Global Forest Watch	Where tree cover was lost, by year	30 m, yearly
Land cover	Dynamic World	Forest, water, crop, built-up per pixel	10 m

Data	Source	What it gives us	Detail
Temperature, soil moisture, wind	ERA5-Land, ECMWF / Copernicus	Regional weather conditions	~9 km grid
Rainfall	CHIRPS	Rainfall, used for drought	~5 km grid

Parameters we measure

We measure four groups of parameters. Some apply only to one ecosystem, because forests and mangroves face different threats.

Group	Parameter	Used for
Vegetation	NDVI (plant greenness)	Forest and mangrove
Vegetation	NDWI (surface water)	Mangrove: separates tides and pond conversion from dieback
Vegetation	Tree cover probability	Forest and mangrove
Weather	Rainfall	Mainly forest (drought)
Weather	Temperature	Mainly forest (heat stress, fire)
Weather	Soil moisture	Mainly forest (drought)
Weather	Wind speed	Mangrove (cyclones)
Events	Fire spots nearby	Mainly forest
Events	Forest area lost	Forest and mangrove
Project	Type, age, area	Forest and mangrove

Main threats: forests face fire, drought and clearing; mangroves face cyclones, erosion and conversion to fish or shrimp ponds.

Parameters vs features

Parameters are raw measurements. Features are what the model uses: each parameter compared to that place's own normal for that month.

A raw value alone means nothing. NDVI of 0.61 is healthy for a dry forest but alarming for a dense rainforest. So for each site we compute its typical value for each calendar month from past years, and the feature is the difference from that normal.

One row = one project, one month:

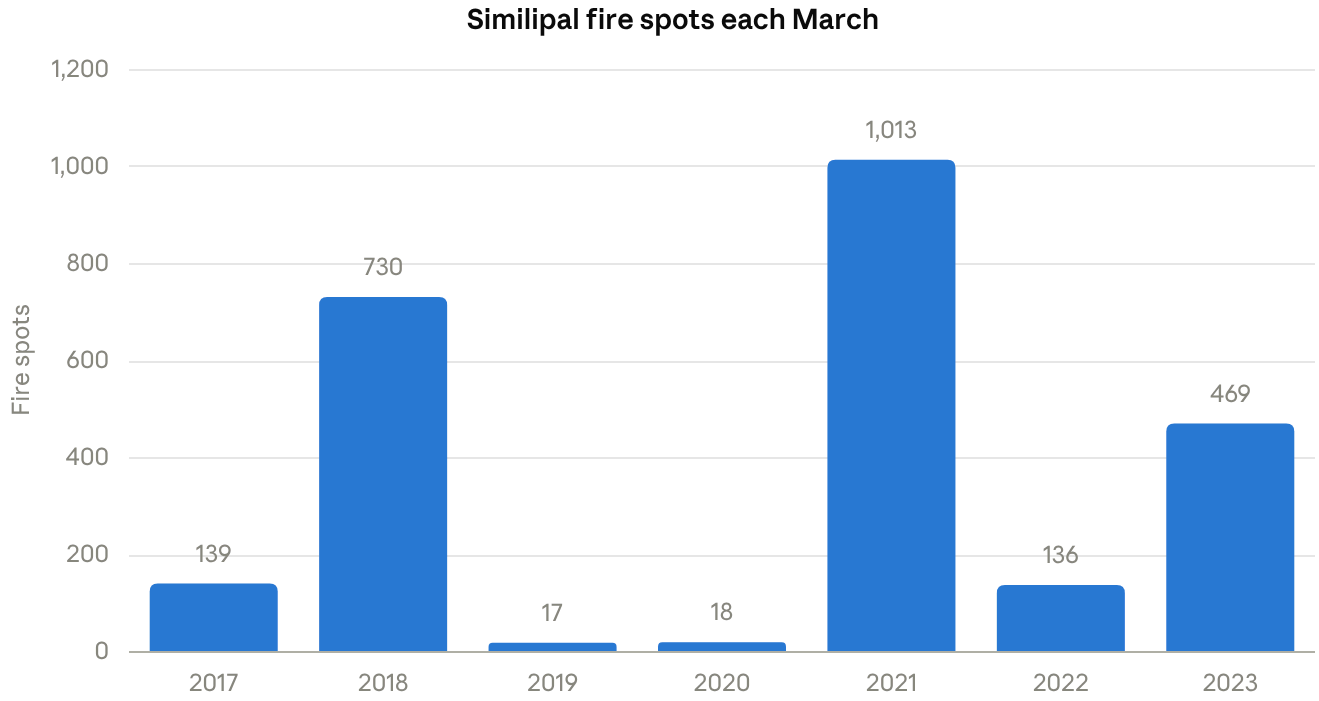
Feature	Example value	Meaning
NDVI anomaly	-0.14	Much less green than a normal April here
Rainfall anomaly (SPI-3)	-1.8	Dry for the last 3 months
Temperature anomaly	+2.1 °C	Hotter than normal
Fire spots within 5 km, last 30 days	3	Recent fire nearby
Tree cover change vs 12 months ago	-0.11	Tree cover falling
Project type, age	Forest, 4 years	Context

The label the model learns from is whether a disturbance followed (forest loss or fire inside the boundary).

Test site: Similipal, Odisha (forest)

Similipal is fixed as our forest test site. Our own code confirms a major fire that peaked in March 2021.

We counted NASA FIRMS fire detections inside a box around Similipal using Earth Engine. In 2021, fire spots rose from 38 in January to 162 in February and 1,013 in March, fell to 102 in April, and stopped once the monsoon arrived.



Fire spots = 1 km cells with a fire detection

Earth Engine FIRMS, our own script · March 2017–2023

March 2021 was about seven times a typical March, so it was truly abnormal. March 2018 gives us a second real event to test on. March 2024 returned 0, which is likely a gap in the Earth Engine copy of the data rather than a fire-free year; we will check it against NASA's FIRMS API. This also shows our system must tell "no fire" apart from "no data".

Decided and still open

Setup is complete and the forest test site is fixed; indicators, thresholds and the mangrove site are decided next.

Item	Status	Next step
Earth Engine, GitHub repo, team setup	Done	Teammates finish their setup
Forest test site: Similipal	Fixed	Check greenness (NDVI) drop and recovery after March 2021
Mangrove test site	Open	Compare Sundarbans (Cyclone Amphan, May 2020) and Coringa, Andhra Pradesh
Indicator list and thresholds per type	Open	Read 3–5 papers per ecosystem

Item	Status	Next step
Label: what counts as "disturbed"	Open	Fix the rule from the same papers
Risk weights	Open	Map to Verra's AFOLU Non-Permanence Risk Tool, then test sensitivity
2024 fire data gap	Open	Cross-check with the NASA FIRMS API

Where something is open, we say when and how we will decide it rather than guessing.